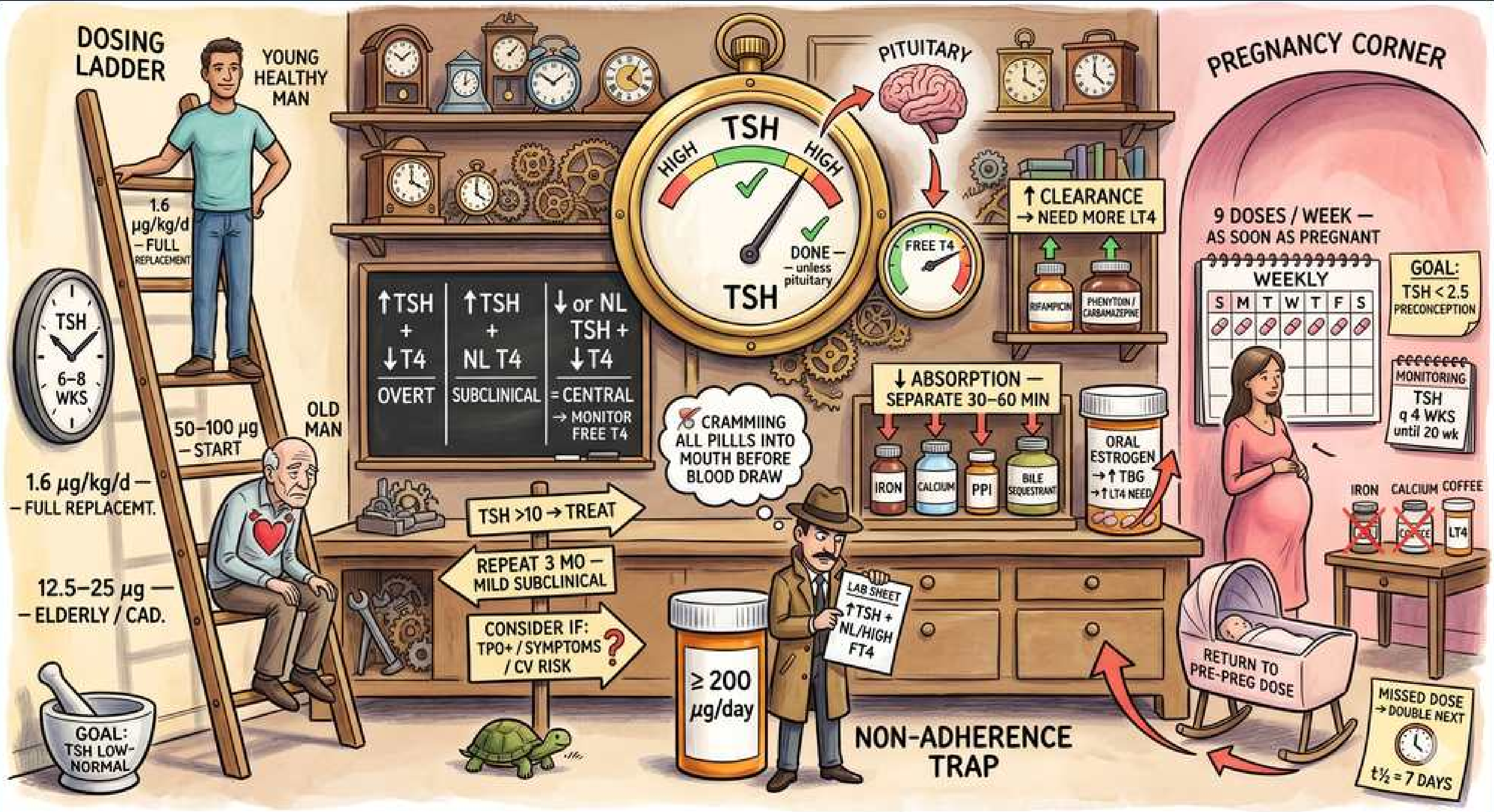
Thyroid — Levothyroxine (LT4) Therapy



1. Full replacement dose

WHAT YOU SEE

'1.6 µg/kg/d — FULL REPLACEMENT', young man on ladder

WHAT IT ENCODES

Full levothyroxine (synthetic T4) replacement is ~1.6 µg/kg/day of ideal body weight in young, healthy adults with overt hypothyroidism. T4 is preferred over T3/combination because of its long half-life (~7 days) giving stable levels and reliable peripheral conversion to T3. Take consistently with the same formulation.

BOARD TRAP

Using LT3 (liothyronine) or T4/T3 combination as routine first-line — wrong; LT4 monotherapy is standard. Combination therapy is not proven superior and gives unstable T3 levels.

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2. Start low in elderly/cardiac

WHAT YOU SEE

A plodding tortoise beside a stooped old man labelled '12.5–25 µg START — ELDERLY/CAD' — the slow tortoise is the 'start low, go slow' titration that protects a fragile heart.

WHAT IT ENCODES

In the elderly or patients with coronary artery disease, START LOW (12.5–25 µg/day) and titrate slowly — abruptly restoring metabolism increases myocardial oxygen demand and can precipitate angina, MI, or arrhythmia. 'Start low, go slow.'

BOARD TRAP

Giving a full weight-based dose to an elderly/cardiac patient — wrong; the surge in metabolic and cardiac demand can trigger ischemia or arrhythmia. Low start, slow up-titration.

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3. Recheck TSH in 6–8 wks

WHAT YOU SEE

A wall of pendulum clocks marked '6–8 WKS' — the workshop full of clocks is the waiting time you must let pass before rechecking TSH after any dose change.

WHAT IT ENCODES

Recheck TSH ~6–8 weeks after any dose change or initiation because LT4's ~7-day half-life means steady state takes ~5–6 half-lives, and TSH lags further. Titrate by ~12.5–25 µg increments based on that TSH.

BOARD TRAP

Rechecking TSH too early (e.g., 2–3 weeks) — wrong; steady state isn't reached and TSH lags, so you'll over- or under-adjust the dose. Wait 6–8 weeks.

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4. TSH titration target

WHAT YOU SEE

Big clock 'TSH HIGH/LOW', goal 'TSH LOW-NORMAL'

WHAT IT ENCODES

In primary hypothyroidism, titrate LT4 to normalize TSH into the reference range (target roughly low-to-mid normal in most; ~0.5–2.5 mIU/L for younger patients, with a higher target acceptable in the elderly). TSH is the monitoring test because the axis is intact.

BOARD TRAP

Aiming for a suppressed TSH in routine replacement — wrong; over-replacement (low TSH) risks atrial fibrillation and osteoporosis. Suppression below normal is reserved for thyroid cancer, not for ordinary hypothyroidism.

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5. Central → follow free T4

WHAT YOU SEE

'↓ or NL TSH → CENTRAL, MONITOR FREE T4'

WHAT IT ENCODES

In central (pituitary/hypothalamic) hypothyroidism, TSH is unreliable (low or inappropriately normal from the lesion), so titrate LT4 to a free T4 in the upper half of the reference range. Always confirm/treat adrenal insufficiency (give glucocorticoid) BEFORE starting LT4.

BOARD TRAP

Titrating central hypothyroidism by TSH — wrong; the broken axis makes TSH uninterpretable, so monitor free T4 instead. Also: starting LT4 before cortisol in coexisting adrenal insufficiency can precipitate adrenal crisis.

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6. Take LT4 on empty stomach

WHAT YOU SEE

A panel '↓ABSORPTION — SEPARATE 30–60 min' showing pills going into the mouth before the blood draw — LT4 must be swallowed fasting, well before food, or absorption drops.

WHAT IT ENCODES

LT4 is absorbed in the small intestine and requires gastric acid; take it FASTING, 30–60 min before breakfast (or at bedtime ≥3 h after the last meal), and consistently. Food (especially coffee, soy, fiber) and many drugs blunt absorption.

BOARD TRAP

Taking LT4 with breakfast/coffee or with other pills — wrong; impaired/variable absorption causes a persistently high TSH that gets misread as needing a dose increase. Fix the timing first.

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7. Separate from binders

WHAT YOU SEE

A cluster of bottles drawn together — IRON, CALCIUM, a PPI, and a BILE-ACID resin — the pile-up of binders that grabs LT4 in the gut, so they must be spaced ~4 hours apart.

WHAT IT ENCODES

Iron and calcium salts chelate LT4; bile-acid sequestrants (cholestyramine) bind it; PPIs/antacids and sucralfate reduce the gastric acid needed for absorption. Separate these from LT4 by at least 4 hours.

BOARD TRAP

Co-administering LT4 with calcium/iron supplements, antacids, or cholestyramine — wrong; absorption drops and TSH rises spuriously. Separate timing by ~4 hours rather than escalating the dose.

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8. Oral estrogen ↑ requirement

WHAT YOU SEE
'ORAL ESTROGEN' ↑ TBG

WHAT IT ENCODES
Oral estrogen (OCP, HRT) and pregnancy raise thyroxine-binding globulin (TBG), increasing bound T4 and lowering free T4 → the patient needs MORE LT4 to keep free hormone adequate. Recheck TSH ~6–8 weeks after starting/stopping estrogen. (Transdermal estrogen has less effect.)

BOARD TRAP
Keeping the LT4 dose unchanged after starting oral estrogen — wrong; the TBG rise increases requirement, so TSH will climb unless the dose is adjusted.

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9. Subclinical: treat if TSH ≥10

WHAT YOU SEE
'TSH >10 → TREAT'

WHAT IT ENCODES
Subclinical hypothyroidism = elevated TSH with NORMAL free T4. Treat unconditionally when TSH ≥10 mIU/L (highest risk of progression to overt disease and adverse cardiovascular outcomes).

BOARD TRAP
Treating every subclinical case regardless of TSH — wrong; the unconditional threshold is TSH ≥10. Below that, treatment is individualized.

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10. Mild subclinical: consider/repeat

WHAT YOU SEE
'REPEAT 3 MO MILD, CONSIDER IF TPO+/SYMPTOMS/CV RISK'

WHAT IT ENCODES
For TSH 4.5–10 (mild subclinical), first REPEAT in ~2–3 months (many normalize spontaneously). Consider treating if symptomatic, anti-TPO positive (higher progression risk), goiter, pregnant/planning pregnancy, infertility, or significant CV risk — especially in younger patients. In the elderly (>65–70), tolerate a mildly higher TSH and avoid over-treatment.

BOARD TRAP
Treating a one-off mildly elevated TSH — wrong; confirm with a repeat first, and weigh age (avoid treating mild elevations in the elderly, where over-replacement risk outweighs benefit).

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11. Pregnancy: ↑ dose early

WHAT YOU SEE
'↑CLEARANCE — NEED MORE LT4', pregnant figure

WHAT IT ENCODES
In pregnancy, increased TBG (estrogen), increased plasma volume, placental deiodinase, and transfer to the fetus raise LT4 requirement by ~25–30%. As soon as pregnancy is confirmed, empirically increase the dose (a common rule: take 2 extra tablets per week → 9 doses/week). Adequate maternal thyroid hormone is essential for fetal neurodevelopment in the first trimester.

BOARD TRAP
Keeping the pre-pregnancy dose unchanged once pregnant — wrong; failing to increase early risks maternal hypothyroidism and impaired fetal neurodevelopment. Increase immediately, don't wait for the next labs.

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12. Pregnancy TSH goals

WHAT YOU SEE
'GOAL: TSH <2.5 PRECONCEPTION; TSH 4–6 WKS'

WHAT IT ENCODES
Target TSH <2.5 mIU/L preconception and in early pregnancy (use trimester-specific ranges), with TSH monitored frequently — every ~4 weeks through the first half of pregnancy and after each dose change. Return to the pre-pregnancy dose immediately postpartum and recheck in ~6 weeks.

BOARD TRAP
Using non-pregnant TSH targets or infrequent monitoring in pregnancy — wrong; pregnancy needs tighter targets (<2.5) and ~4-weekly checks. Also remember to drop back to the pre-pregnancy dose after delivery to avoid over-replacement.

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13. Non-adherence trap

WHAT YOU SEE
'NON-ADHERENCE TRAP — ≥200 µg/day', cramming pills

WHAT IT ENCODES
A patient requiring an unusually high dose (e.g., ≥200 µg/day or >2 µg/kg) with persistently abnormal/erratic labs is far more likely non-adherent (or has absorption/timing problems) than truly malabsorptive. A levothyroxine absorption test (supervised loading dose with serial T4 levels) can confirm normal absorption and unmask adherence issues.

BOARD TRAP
Escalating the dose ever-higher for suspected non-adherence or 'malabsorption' — wrong; once the patient takes it consistently they become thyrotoxic. Investigate adherence/timing and absorption before increasing.

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14. Missed dose → double next

WHAT YOU SEE
'MISSED DOSE → DOUBLE NEXT'

WHAT IT ENCODES
Because LT4 has a long ~7-day half-life, a missed dose can be made up by taking two tablets the next day (or extra doses spread over the week) without causing toxicity — total weekly dose is what matters for steady levels.

BOARD TRAP
Telling patients to simply skip a missed dose with no make-up — the long half-life makes catch-up dosing safe and appropriate; chronic skipping under-replaces them.
